

RUHR-UNIVERSITÄT BOCHUM

Countermeasures against BGP hijacking

Sabri Öztürk

Seminar on Security and Privacy of Ubiquitous Systems

Seminar on Security and Privacy of Ubiquitous Systems

May 13, 2026

Chair for Security and Privacy of Ubiquitous Systems

Supervisor: [redacted]



Contents

1	Introduction	7
2	Technical Background	9
3	Attacks on BGP	11
4	Countermeasures against BGP-Hijacking	13
4.1	IETF-Standards	13
4.2	DISCO	15
4.3	ROV++	16
5	Results	19
6	Discussion	23
7	Conclusion	25
	Bibliography	27

Abstract

The Border Gateway Protocol (BGP) is a trust-based routing protocol that facilitates the exchange of network routing information between autonomous systems. However, it is vulnerable to attacks, because announcing routing information as an autonomous system does not require authentication or proof of de facto ownership of IP addresses. Researchers found that over 90% of Tor relays are vulnerable to BGP hijacking attacks [SEV⁺15]. This paper will summarise the existing countermeasures against BGP hijacking and try to compare them. We will see that the standard recommendation, ROV with RPKI, suffers due to its adaptation problems. In our evaluation, ROV++ offers additional choices that provide more security against hijacks without significantly reducing deployability. Meanwhile, DISCO stands out with its automation of ownership verification, making the deployment of RPKI easier. Finally, we will discuss what our results mean, what countermeasures are recommended, and what areas should be researched in the future.

Abbreviation

BGP: Border Gateway Protocol

AS: Autonom-System

ASes: Autonom-Systems

ASN: Autonom-System-Number

IETF: Internet Engineering Task Force

TCP: Transmission Control Protocol

IP: Internet Protocol

ROV: Route Origin Validation

ROV++: Route Origin Validation++

RPKI: Resource Public Key Infrastructure

RAPTOR: Routing Attacks on Privacy in Tor

DISCO: Decentralized Infrastructure for Securing and Certifying Origins

ISP: Internet Service Provider

NLRI: Network Layer Reachability Information

ROA: Route Origin Authorization

RC: Resource Certificate

1 Introduction

Motivation BGP is a trust-based, inter-domain routing protocol used to exchange information about network routing between autonomous systems. BGP sessions are established over TCP connections (typically on port 179) to ensure reliable transmission and avoid packet loss during route advertisement. [RLH06]

However, BGP lacks security mechanisms. It implicitly trusts the routing information it receives from peers. This makes the protocol vulnerable to BGP hijacking, where malicious ASes can manipulate internet routing. Detecting the hijack is made more difficult by BGP interception, where attacker forwards the hijacked internet traffic to its legitimate destination after intercepting it.

An AS-level attacker uses BGP hijacking to perform internet routing over their own AS. While it may sound safe to use encryption to hide content and MACs to detect content manipulation, which secures against eavesdropping, the adversary can still carry out various attacks, such as blackholing. Blackholing prevents internet traffic from reaching its destination, disconnecting users from the Internet. Other possible attacks will be discussed later.

Goals and Contribution This paper will summarize some of the existing countermeasures for BGP hijacking. Then we will evaluate the efficiency of their implementation, usage and security effectiveness.

Structure Firstly, we will provide some basic information about BGP and Tor. We will define important terms.

Then we will introduce Hijack-Attacks on BGP.

Next, we will briefly explain the current state of countermeasures and discuss their problems.

Then we will look at modern countermeasures: DISCO and ROV++.

Finally, we will present comparative results of IETS standards, DISCO and ROV++.

2 Technical Background

BGP BGP-4 is the current version of BGP. iBGP (internal BGP) is used between routers within the same AS. eBGP (external BGP) is used between routers that belong to different ASes. When we talk about an attacker, we mean an AS-level attacker. This is because BGP hijacking is mostly carried out via eBGP. Therefore, when we talk about BGP/BGP hijacking, we specifically refer to eBGP/eBGP hijacking. An ISP is an organisation, that provides access to the internet. They are responsible for establishing a physical connection to the internet and assigning IP addresses to their users. Every ISP requires an AS in order to route traffic and connect its users to the global internet. Each AS is identified by a unique ASN. The BGP protocol is used to build routing tables. ASes connect to each other via a peer-to-peer connection using a TCP (port 179) to prevent information loss. There are four different BGP message types:

- 1. Open: establishes a BGP session.
- 2. Update: Used for exchanging routing information. There are three main optional sections: Routing Attributes; Network Layer Reachability Information (NLRI), which includes prefixes to be added; and Withdrawn Routes, which includes prefixes to be removed. These sections provide enough information for inter-AS routing decisions.
- 3. Keepalive: responsible for session refresh.
- 4. Notification: sends notifications about errors and closes the session. [RLH06]

BGP is trust-based: ASes trust update messages from peers and add announced IP prefixes to their routing tables unless external implementations apply validations [RLH06]. IP prefixes and AS paths are exchanged between ASes. ASes share these IP prefixes with their neighbors, creating an iterative process. Instead of sending the full routing table each time, BGP only advertises new or withdrawn routes, minimizing overhead. BGP uses MAC MD5 [RLH06] to ensure message integrity and authenticity. However, MD5 is now considered weak and outdated due to known security vulnerabilities.

Routing Decisions There are many possible paths for IP packets to travel from source A to destination B. The specific path taken must be determined by routers. In general, routing decisions can be configured individually. However, there are two main criteria that most routers follow [HCG⁺20].

First, the shortest AS path is preferred. ASes typically choose the route with the fewest number of AS hops, even if they have already received the original IP prefix from its legitimate owner. If the same prefix is received via a shorter route, that route is prioritized.

In BGP, routers prefer the most specific prefix, meaning the longest bit sequence that matches the destination IP address. For example, assume that AS1 announces the prefix 1.1.0.0/20 and AS2 announces 1.1.0.0/21. Both prefixes cover the same destination IP address (e.g., 1.1.0.5). In this case, routers will prefer the route through AS2, because 1.1.0.0/21 is more specific than 1.1.0.0/20.

Tor The Tor network is a system for anonymous communication. The Internet Protocol (IP) by itself does not provide anonymity. Even if the content of a message is encrypted, an eavesdropper can still see the IP addresses of both the sender and the receiver. Using certain methods, an IP address can be used to determine the physical location of the user or collect other metadata that may reveal sensitive information about the parties involved in the communication. To eavesdrop on the communication, it is enough to monitor just one hop that the communication passes through.

Tor solves this problem by routing traffic through multiple relays (nodes) and applying layered encryption to both the payload and the headers at each hop. Encryption is used to ensure that each relay learns the identity of only the previous hop and the next hop in the communications, and no single relay can link the client to the destination server. The entry (guard) relay knows the user's IP address but not the final destination, while the last (exit) relay knows the destination but not the source. This way, Tor effectively hides the IP headers at each step, preventing direct tracking of users.

3 Attacks on BGP

BGP hijacking is an active man-in-the-middle attack that reroutes Internet traffic. The goal of these attacks is always to insert the attacker's own connection into the traffic flow. Through BGP hijacking, the attacker receives Internet packets, which is just the first step enabling a variety of further attacks. We will discuss some of these attacks on this section, after introducing some categories of BGP hijacking.

Prefix hijacking AS-level adversaries can hijack an IP prefix by announcing the prefix as its own. The attack causes a fraction of Internet traffic destined to the prefix to be captured by the adversary. [SEV⁺15]

Subprefix hijacking The hijacking AS announces routes to a more specific prefix address than the victim prefix (e.g., /24 instead of /16). Because of the longest prefix match rule, any router that receives the incoming route without filtering it will automatically forward any traffic destined to that subprefix to the hijacking AS. [Jac15]

Superprefix hijacking An attacker announces a less specific prefix (superprefix) that includes a more specific, non-routed subprefix. Since the subprefix was never advertised, routers accept the superprefix. For example, suppose the prefix 1.2.3.0/24 is non-routed, but belongs to AS 0. An attacker, AS 6, aims to hijack 1.2.3.0/24 and therefore announces the superprefix 1.2.0.0/22. As a result, AS 6 will successfully hijack almost all traffic sent to 1.2.3/24. [MFM⁺21]

If an attacker uses a non-routed prefix to carry out a hijacking, it is called a non-routed attack. For example, superprefix hijacking is one such attack.

AS-Path hijacking Instead of claiming ownership of an IP prefix, the attacker falsifies the AS path by pretending to have a valid route to the destination AS. AS A tells AS B 'I can be bridge for your routing to AS C.' [SCB13]

By hijacking a prefix, an attacker may [ZJP⁺07]

- create a blackhole in the network (by simply dropping the packets);
- steal the victim AS's identity by imitating the victim's services (e.g. duplicate website);
- intercept the traffic to eavesdrop (or record) the exchanged data and then forward the data back to the victim AS;
- send spam, phishing, and Denial-of-Service traffic [MFM⁺21]

RAPTOR Tor is known to be vulnerable to attackers who can observe traffic at both ends of the communication path. Routing Attacks on Privacy in Tor are traffic correlation attacks that exploits this vulnerability to compromise user anonymity in the Tor network. [SEV⁺15] There are three main techniques to initiate RAPTOR attacks, but in this work, we focus specifically on the variant that uses BGP hijacking.

This type of RAPTOR attack can be launched by AS-level adversaries that hijack specific IP prefixes along the client-to-entry and exit-to-destination paths of a Tor circuit. This type of RAPTOR attack was demonstrated in a simulated setting where AS-level adversaries hijacked specific IP prefixes along Tor circuits. By collecting TCP ACKs on both the client and server sides, the attacker can correlate them based on matching timing and data-volume patterns. This correlation allows linking traffic flows, thereby linking users. Linking users effectively breaks anonymity [LJP10]. In their experimental setup, the attackers achieved up to 90% accuracy in deanonymizing users. [SEV⁺15]

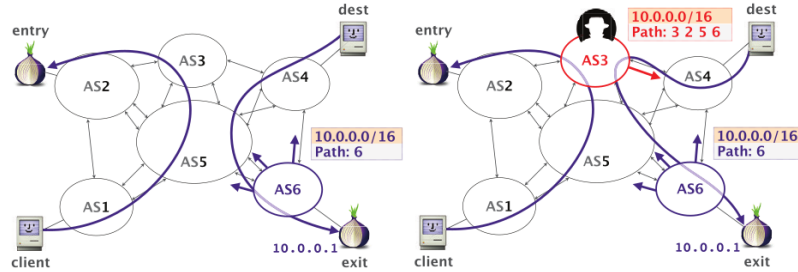


Figure 3.1: Raptor attack via BGP hijacking[SEV⁺15].

Figure 4.2 illustrates a malicious AS3 initiating a BGP hijacking attack and forwarding the packets to the destination, thereby achieving interception. By observing TCP ACKs on both the client and server sides, the attacker can correlate them based on matching timing and data volume patterns, effectively linking users.

4 Countermeasures against BGP-Hijacking

BGP suffers from the fact that announcing routing information as an AS does not require authentication or proof of actual ownership of IP addresses. One radical solution to prevent BGP hijacking would be to require such verification before an AS is allowed to participate or send any BGP messages. While this could effectively eliminate the hijacking problem, it would cause significant availability issues. Since the core philosophy of the Internet prioritizes availability over security, imposing strict participation requirements is not a viable solution to address the problem at its root. Therefore, it is essential to explore and develop external security frameworks that enhance BGP security without compromising the fundamental principle of network availability.

The Internet Engineering Task Force (IETF) develops internet standards and is the official body responsible for the BGP standard [RLH06]. Therefore, we will first examine RPKI and BGPsec, which are IETF-standardized countermeasures against BGP hijacking. Then, we will discuss their limitations and introduce DISCO and ROV++, two approaches that aim to support and evaluate RPKI in addressing its problems.

4.1 IETF-Standards

Here, we will briefly explain the current state of affairs and the remaining problems without delving too deeply into technical details.

Resource Public Key Infrastructure(RPKI) RPKI is a public key infrastructure designed to secure the Internet’s routing system. It allows legal IP prefix owners to create cryptographically signed ROAs, which can be published in public repositories and validated by routers (ROV). All other countermeasures we will introduce depend on it or complement it. There are three main components (RC, ROA and ROV) that RPKI uses, which we will define next [HCG⁺20].

Resource Certificate (RC) A Resource Certificate binds an IP prefix to the prefix owner’s public key, verifying their right to use the resource. It serves as cryptographic proof of legitimate ownership within the RPKI system and is required to create a ROA [HCG⁺20].

Route Origin Authorization (ROA) ROA is a cryptographically signed document in the RPKI system that authorizes an AS to announce IP prefixes via BGP. It can be seen as a signature that binds an IP prefix to its origin AS.

It includes a `maxLength` parameter, which defines how specific (i.e., long) the announced IP prefix may be. This serves as a countermeasure against subprefix hijacking. However, its effectiveness depends on the deployment of Validation (e.g. ROV/ROV++) [HCG⁺20].

Route Origin Validation (ROV) ROV checks whether a BGP announcement matches the information in a ROA. Based on the result, it accepts, ignores, or rejects the announcement. Routers then apply the corresponding changes to the routing table. If ROV rejects, no changes are made. If ROV accepts, changes are applied. If ROV ignores, the decision is taken without considering the ROA. If a legitimate prefix has a ROA in the RPKI repository, any conflicting announcement for this prefix would be rejected [MFM⁺21].

BGPsec is an extension to BGP that secures the AS path in BGP UPDATE messages using digital signatures. Each AS that propagates the message adds its own signature. These digital signatures provide confidence that every AS in the path listed in the UPDATE message has explicitly authorized the advertisement of the route [LS17]. This extension prevents against AS-path hijacking. However, the adoption of BGPsec is far more difficult than that of RPKI, as it requires changes to the Internet infrastructure to support on-path cryptography [HCG⁺20].

IETF standards are not mandatory. ASes are not forced to implement them. Instead, these standards can be seen as strong recommendations or best practices. In summary, RPKI binds IP prefixes to the owner’s public key and is used to prove ownership of resources. However, as RPKI must be deployed manually by ASes, adoption is difficult and prone to misconfiguration, since there are many complex and manual steps in RPKI [HCG⁺20]. ROV is primarily designed to prevent prefix hijacking, which is considered the most critical attack on BGP. Announcements with no ROA are treated as not found rather than being dropped. The decision is then made without considering the ROA, as mentioned before. That’s why prefix hijacking can still happen in cases where announcements with no ROA are accepted.

One of the main problems is the low deployment rate of RPKI and ROV. This is due to the complexity of certifying IP prefix ownership, a process that is manual and bureaucratic. Another major issue is misconfiguration: Thirty percent of ROAs are misconfigured. Even worse, over 5% of ROAs conflict with legitimate long-lived BGP announcements, causing enforcing routers to mistakenly drop valid routes and disconnect users [HCG⁺20].

Together, these security and deployability challenges limit the practical effectiveness of RPKI. To address these challenges, we introduce DISCO, which aims to simplify RPKI implementation and reduce misconfiguration. We also present ROV++, which improves upon ROV's protection against hijacking. ROV++ secures against hijackings and is more effective than ROV at low deployment levels.

4.2 DISCO

DISCO is a system that certifies ownership of IP address blocks. It provides substantial security benefits and circumvents the adoption obstacles faced by RPKI and ROV [HCG⁺20]. Unlike the bureaucratic and problematic authentication process in RPKI, DISCO automatically proves prefix ownership. It is an automated routing certification system that delivers easy-to-use, error-free ROAs.

This mechanism operates as follows DISCO registers validate pairs of prefixes and public keys by applying following steps 4.1. DISCO requires ASes to include Public Keys (PKs) in their BGP announcements. By sharing a BGP announcement, the PKs remains included. Vantage points, which are locations in the Internet where routing behavior can be observed, collect BGP announcements. The DISCO Registers receive these BGP announcements. If the same public key is received from many different vantage points, the register applies it as proof of de facto IP prefix and sends its partial signature, or 'share', to the repository. Once enough Registers agree, meaning that a threshold of partial shares has been reached in the repository, the public key becomes certified, and this is called a Routing Certificate. These routing certificates are sent to other repositories for availability. DISCO RCs complement RPKI RCs and are used just like them. Everyone can download them.

ROAs are issued automatically by DISCO. Since DISCO holds the private signing key, it can generate ROAs on demand for each (sub)prefix, specifying the MaxLength attribute to cover all approved subprefixes. The DISCO agent monitors repositories and detects when a new Resource Certificate (RC) has been published. It then uses its offline signing key—either automatically or semi-automatically—to sign a ROA for each (sub)prefix authorized by the AS. These ROAs are published to RPKI repositories and retrieved by validators, enabling routers to apply them without manual intervention, effectively allowing DISCO to certify de facto 4.2.

DISCO does not support BGPsec, instead, it is well-suited for path-end validation, an alternative to BGPsec that works without infrastructure changes and supports partial deployment. In this approach, a prefix owner approves neighboring ASes using its private key. Since DISCO maps IP addresses to public keys, it naturally supports verifying these approvals and helps defend against path manipulation attacks.

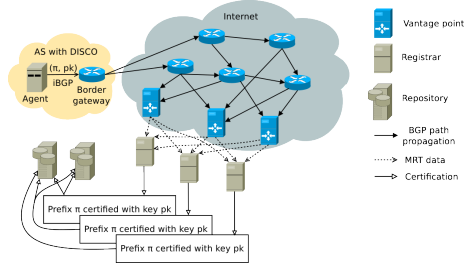


Figure 4.1: Certification through DISCO. DISCO agent binds public key with IP prefix [HCG⁺20].

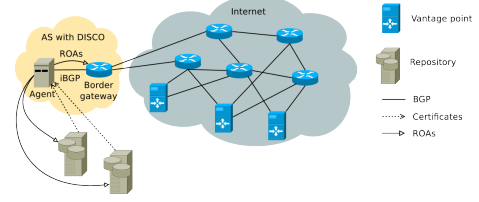


Figure 4.2: Origin validation through DISCO. DISCO agent issues a ROA, and fetches ROAs issued by others [HCG⁺20].

Is it decentralized and secure? Since many registers are involved, we don't need to trust individual ones. The redundancy of registers makes the system decentralised, thereby improving security. Repositories are signed, so they cannot be tampered with. To ensure availability, we use redundant repositories.

4.3 ROV++

ROV++ is an extension of ROV to improve its security. ROV++ extends ROV by incorporating various mechanisms with different impacts and trade-offs. ROV++ reduces subprefix hijack rates, particularly in cases of low adoption. ROV prevents simple prefix-hijacks, but fails when an upstream AS is hijacked: even if downstream ASes are not hijacked, they inherit the wrong route. ROV also can't defend against superprefix hijacking, for example, when an attacker announces 192.0.2.0/24 to steal traffic for a more specific but never-advertised 192.0.2.128/25. ROV++ addresses these gaps with three complementary defences. ROV++ introduces three key defense mechanisms [MFM⁺21]:

- **Visible Hijack Prevention:** When an invalid prefix is observed and no valid alternative route exists, ROV++ blackholes (drops) the hijacked route to prevent misrouting.
- **Blackhole Announcement Sharing:** ROV++ routers not only drop invalid prefixes but also announce special blackhole BGP messages. This allows neighboring ROV++ ASes to identify and also drop traffic to these invalid routes.
- **Preventive Subprefix Re-announcement:** Upon detecting a hijack, if both a legitimate and a forged announcement for the same subprefix exist, the ROV++ router actively re-announces the correct subprefix along the valid AS path to steer traffic back to the legitimate origin.

Based on these defenses, ROV++ has 5 variants.

- **ROV++ v1:** Implements only *Visible Hijack Prevention*. It is recommended to implement it against routed prefix hijacks [MFM⁺21].
- **ROV++ v2:** Combines *Visible Hijack Prevention* and *Blackhole Announcement Sharing*. It offers stronger protection but may blackhole some legitimate traffic. ROV++ v2 (or a variant) for non-routed prefix hijacks [MFM⁺21].
- **ROV++ v2a:** ROV++ v2a builds on v2 by aggressively blackholing hijacks, particularly for non-routed prefixes. Upon receiving an invalid announcement from any neighbour, it immediately sends blackhole announcements to its peers.
- **ROV++ v3:** Implements all three mechanisms, including *Preventive Subprefix Re-announcement*. It offers the strongest defense in complex hijack scenarios, but also introduces the most operational overhead. It does not perform better than v1 against routed hijacking, nor better than v2 against non-routed hijacking, as it carries a risk of abuse by attackers. Therefore, it is not recommended to implement it.
- **ROV++ Lite-Versions:** Lite variants of each versions are to be preferred for their ease of deployment, as they have minimal impact on security effectiveness.

Each variant is designed with different trade-offs between deployability, aggressiveness, and defense coverage, allowing network operators to choose the most appropriate version based on their infrastructure and risk model. ROV cannot secure against subprefix attacks if the entire path to the destination does not use ROV. For example, an upstream AS, 44, is hijacked by AS 666, which offers the shortest AS path to reach the non-hijacked AS 99. AS 666 announces a more specific prefix, and AS 44 routes to AS 666. AS 77 and AS 78 detect the hijack by AS 666 and drop that announcement. However, they still use AS 44 to reach AS 99, as it is the shortest path. But AS 44 routes these packets to AS 666 4.2(a). ROV++ (Visible Hijack Prevention) blackholes if no non-hijacked path is found, or uses a path to reach the destination if a non-hijacked path is available. Thus, no hijacking takes place and availability is preserved, if possible 4.2(b).

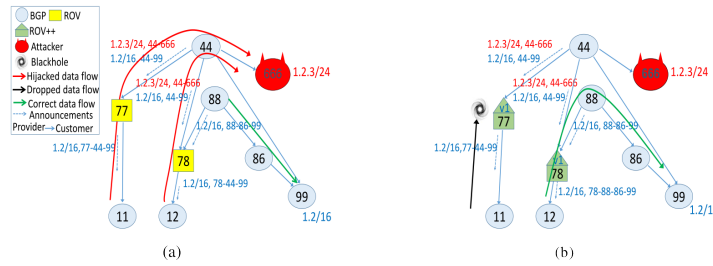


Figure 4.3: (a) ROV on ASes 77, 78 is not enough to prevent subprefix hijacking. (b) ROV++v1 blackholes if no non-hijacked path found or uses a path to reach the destination, if a non-hijacked path found [MFM⁺21].

5 Results

RAPTOR highlights the importance of securing internet routing, particularly the AS-Path. If your packets pass through malicious ASes, you can only ensure their confidentiality and integrity using encryption and MACs, but you cannot guarantee their availability or prevent them from being used to deanonymize you.

We selected these countermeasures because IETF standards are generally the ones companies rely on the most. ROV++ and DISCO have emerged as company-friendly, easy-to-deploy solutions with no extra requirements, minimal risk of errors, and a 'do no harm' principle. Overall, they effectively address the security and deployment gaps of RPKI/ROV. For the "Feature" section, we focused on categories of BGP hijacking attacks as well as key requirements relevant to adoption feasibility. Here, we compare the countermeasures based on their features.

Feature / Countermeasure	ROV	BGPsec	DISCO	ROV++ v1	ROV++ v2	ROV++ v3	ROV++ Lite
Prevents against Prefix hijacking	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Prevents against Subprefix hijacking	Partial	Yes	Mostly	Yes	Yes	Yes	Yes
Prevents against Superprefix hijacking	No	Yes	No	No	Yes	Yes	Yes
Prevents against AS-Path hijacking	No	Yes	No	No	No	No	No
Supports RPKI infrastructure scaling	No	No	Yes	No	No	No	No
Deployment Complexity	Low	Very High	Medium	Medium	Medium	Medium	Low
Operational Overhead	Low	High	Low	Low	Medium	Medium	Low
Security Effectiveness with low adaptation	Low	Low	Low	Medium	High	Very High	Medium/High
Implementation Recommendation	Very High	Low	High	Medium	Medium	Low	High
Extra Requirements		Per-hop path signing	PK on BGP-Announcements				

Operational overhead is rated independently of the RPKI infrastructure requirement. Although DISCO utilises RPKI, it does so in a more flexible and streamlined manner than others. Therefore, its operational overheads are technically much lower.

The ratings in this table are based on a qualitative comparison of the mechanisms discussed in Section 5, using the following criteria: deployability (real-world adoption feasibility), security coverage (types of hijacks mitigated), and operational complexity. These were derived from the original proposals [MFM⁺21, HCG⁺20].

Reasoning Ratings In this section, we explain the ratings. We rated BGPsec’s security effectiveness (highest 4 features) assuming it is applied to the whole Internet. If it is only partially deployed, it does not provide better results than RPKI/ROV [MFM⁺21].

Prefix hijacking is prevented by every countermeasure, since it is considered the biggest problem of BGP. It is actually the motivation behind RPKI/ROV, which can be seen as the first serious step for BGP security.

Subprefix hijacking is partially prevented by ROV, but fully prevented by all variants of ROV++ thanks to its Visible Hijack Prevention algorithm, as we explained in the ROV++ section. For ROV to be effective against data-plane subprefix hijacks, it requires an extremely widespread deployment, which is not expected in the foreseeable future [MFM⁺21]. As DISCO relies on ROV, it cannot prevent attacks completely. Correctly configured ROAs provided by DISCO help ROV perform well, but ROV’s limitation discussed in 4.2 would not be solved. Correctly configured ROAs also help detect subprefix hijacking attempts, which is why we rate it higher than ROV.

AS-Path hijacking can only be prevented by BGPsec, since it signs and verifies the full AS path at every hop. All other countermeasures do not address path manipulation. DISCO helps to evaluate an alternative approach that supports authenticating neighbors through path-end validation records, but it does not apply it.

Only DISCO supports RPKI scaling, as it automates the creation and distribution of Routing Certificates and ROAs.

Deployment complexity: is low for ROV++ Lite, since it is software-only, readily-deployable design [MFM⁺21]. It is also low for ROV, as it is requirement for all other countermeasures, it cannot be harder than any other. Deploying BGPsec is by far the hardest, as we discussed in the BGPsec section.

ROV++ v2 and v3 introduce additional logic and blackhole routes, increasing management burden. DISCO and Lite variants remain lightweight.

The more ROAs are deployed, the more effective their security becomes [MFM⁺21]. Although effectiveness with low adoption increases for ROV++ variants as they gain additional mechanisms, their effect on non-ROV-implemented ASes also increases. The effectiveness of ROV++ Lite depends on which variant it is based on; it mirrors the variant it implements. DISCO can strengthen effectiveness if widely adopted and integrated with existing validation systems.

ROV is hardly recommended for implementation, since without it there is no way to validate ROAs automatically. DISCO is highly recommended, as it complements RPKI and solves its problems without requiring too much effort. ROV++ v1 and ROV++ v2 are not strongly recommended, since their Lite versions are much easier to adopt without significant differences in security effectiveness. BGPsec is not recommended due to the extensive changes it requires across the Internet.

ROV++ Lite (for v1 or v2) has been identified as the most implementation- and security-efficient countermeasure for validation, due to its minimal operational overhead and lack of need for extensive modifications to BGP router logic. DISCO offers an alternative approach that helps overcome the deployment challenges of RPKI. It requires only a public key in BGP announcements, which is easy to implement based on existing Internet standards, as a dedicated section already exists for this purpose [HCG⁺20]. Its support for path-end validation and elimination of manual configuration make it a comprehensive, practical, and effective solution. It is particularly well-suited for simplifying RPKI deployment. BGPsec, on the other hand, may be considered for AS paths with very high security requirements or for those that are used frequently.

As an AS without any security implemented, the first step is to implement RPKI to own secure prefixes. For easier, faster and error-free implementation of RPKI, DISCO should be preferred. Then, deploy ROV for validation. After that, the extension ROV++ Lite is a good addition that improves security effectiveness.

6 Discussion

Biases This paper did not evaluate live BGP route propagation or performance metrics. The given statistics are mostly outdated, as no recent research was found. The rating in the Results section is an interpretation of the available information, used when explicit data is not provided. These interpretations may contain mistakes that could be very misleading. In particular, comparisons between countermeasures such as DISCO and ROV++ are based on understanding, as they have never been tested in the same setup. Combining ROV++ and DISCO is a promising idea, but we could not try it out due to the lack of resources to set up such an environment. We discussed deployment issues only from a technical perspective; the actual reasons why companies do not engage remain unknown.

Future Work All revealed countermeasures could be tested in the real world. The rating could be presented in a graph instead of a table, which would show the differences in more detail. Conducting interviews with companies that configure ASes could provide valuable insight into the reasons behind the low deployment of RPKI/ROV. Additionally, combining DISCO and ROV++ should be explored.

7 Conclusion

We presented different countermeasures against BGP hijacking, discussed their issues, and compared their security effectiveness and adoption feasibility. We presented different ways to carry out BGP hijacking attacks. We presented RAPTOR as it did show a huge rate of success, underlines the need for deployable BGP security. While IETF standards offer initial approaches, manageable and fault-resistant solutions such as DISCO and ROV++ remain critical for widespread adoption and improving security. DISCO prevents hijacking and AS path hijacking by verifying the ownership of IP addresses. It is a complementary, automated routing certification system offering easy-to-use, non-error ROAs and RCs. ROV++ improves upon ROV by offering better statistical protection against hijacking attacks and being easy to deploy.

Bibliography

- [HCG⁺20] Tomas Hlavacek, Italo Cunha, Yossi Gilad, Amir Herzberg, Ethan Katz-Bassett, Michael Schapira, and Haya Shulman. Disco: Sidestepping rpki's deployment barriers. In *Network and Distributed System Security Symposium (NDSS)*, 2020.
- [Jac15] Quentin Jacquemart. *Towards uncovering BGP hijacking attacks*. PhD thesis, Télécom ParisTech, 2015.
- [LJP10] Lucie Langer, Hugo Jonker, and Wolter Pieters. Anonymity and verifiability in voting: understanding (un) linkability. In *Information and Communications Security: 12th International Conference, ICICS 2010, Barcelona, Spain, December 15-17, 2010. Proceedings 12*, pages 296–310. Springer, 2010.
- [LS17] Matthew Lepinski and Kotikalapudi Sriram. Rfc 8205: Bgpsec protocol specification, 2017.
- [MFM⁺21] Reynaldo Morillo, Justin Furuness, Cameron Morris, James Breslin, Amir Herzberg, and Bing Wang. Rov++: Improved deployable defense against bgp hijacking. In *NDSS*, 2021.
- [RLH06] Yakov Rekhter, Tony Li, and Susan Hares. Rfc 4271: A border gateway protocol 4 (bgp-4), 2006.
- [SCB13] Johann Schlamp, Georg Carle, and Ernst W Biersack. A forensic case study on as hijacking: the attacker's perspective. *ACM SIGCOMM Computer Communication Review*, 43(2):5–12, 2013.
- [SEV⁺15] Yixin Sun, Anne Edmundson, Laurent Vanbever, Oscar Li, Jennifer Rexford, Mung Chiang, and Prateek Mittal. {RAPTOR}: Routing attacks on privacy in tor. In *24th USENIX Security Symposium (USENIX Security 15)*, pages 271–286, 2015.

- [ZJP⁺07] Changxi Zheng, Lusheng Ji, Dan Pei, Jia Wang, and Paul Francis. A light-weight distributed scheme for detecting ip prefix hijacks in real-time. *ACM SIGCOMM Computer Communication Review*, 37(4):277–288, 2007.